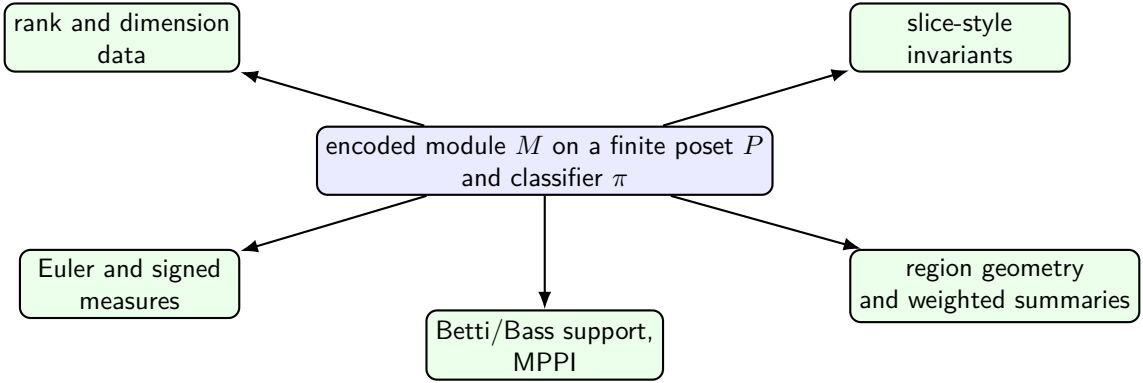
Invariant Families in TAMER-OP

What the library computes after finite encoding

In multiparameter persistence, there are many invariants, not one barcode

For a finitely encoded module, the primary object is still the encoded module itself, not a single complete summary. Different invariants emphasize different aspects:



The code therefore implements a *family* of invariants and geometry-aware summaries, all built from the same finite encoded object.

Complete inventory by family I

Family	Implemented functions
Rank and dimension	`restricted_hilbert`, `rank_map`, `rank_invariant`, `rank_invariant_tame`, `ecc`, `hilbert_distance`
Rectangle and rank inversion	`rectangle_signed_barcode`, `rectangle_signed_barcode_rank`, `rank_from_signed_barcode`, `truncate_signed_barcode`, `rectangle_signed_barcode_image`, `rectangle_signed_barcode_kernel`, `rectangle_measure`, `rectangles`
Slice barcodes	`slice_chain`, `restrict_to_chain`, `slice_barcode`, `slice_barcodes`, `slice_chain_exact_2d`, `fibered_slice`, `fibered_slice_family_2d`
Slice distances	`bottleneck_distance`, `wasserstein_distance`, `sliced_bottleneck_distance`, `sliced_wasserstein_distance`, `sliced_wasserstein_distance_approx`, `matching_distance`, `matching_distance_approx`, `matching_distance_approx_2d`, `matching_distance_exact_2d`, `matching_distance_exact_slices_2d`
Slice features and kernels	`PersistenceLandscape1D`, `persistence_landscape`, `PersistenceImage1D`, `persistence_image`, `persistence_silhouette`, `barcode_entropy`, `barcode_summary`, `slice_features`, `slice_kernel`, `wasserstein_kernel`
Projected and fibered caches	`FiberedArrangement2D`, `FiberedBarcodeCache2D`, `fibered_barcode`, `fibered_barcode_index`, `fibered_values`, `ProjecteArrangement`, `ProjectedBarcodeCache`, `projected_arrangement`, `projected_barcode_cache`, `projected_barcodes`, `projected_distances`, `projected_distance`, `projected_kernel`

Complete inventory by family II

Family	Implemented functions
Euler and signed measures	`euler_characteristic_surface`, `euler_surface`, `euler_signed_measure`, `euler_distance`, `PointSignedMeasure`, `point_signed_measure`, `surface_from_point_signed_measure`, `truncate_point_signed_measure`, `point_signed_measure_kernel`
Multiparameter landscape and image families	`MPLandscape`, `mp_landscape`, `mp_landscape_distance`, `mp_landscape_inner_product`, `mp_landscape_kernel`, `multiparameter_landscape`, `multiparameter_landscape_standard`, `MPPLineSpec`, `MPPDecomposition`, `mpp_decomposition`, `MPPImage`, `mpp_image`, `mpp_image_distance`, `mpp_image_inner_product`, `mpp_image_kernel`, `mma_decomposition`
Weighted module summaries	`integrated_hilbert_mass`, `measure_by_dimension`, `measure_by_value`, `support_measure`, `vertex_set_measure`, `region_values`, `dim_stats`, `dim_norm`, `region_weight_entropy`, `aspect_ratio_stats`, `module_size_summary`, `module_geometry_summary`
Interface and graph summaries	`interface_measure`, `interface_measure_by_dim_pair`, `interface_measure_dim_changes`, `graph_degrees`, `graph_connected_components`, `graph_modularity`, `region_adjacency_graph_stats`, `region_volume_samples_by_dim`, `region_volume_histograms_by_dim`, `region_boundary_to_volume_samples_by_dim`, `region_boundary_to_volume_histograms_by_dim`
Algebraic support descriptors	`betti_support_measures`, `bass_support_measures`, `upsets_covering`, `downsets_covering`, `minimal_primes`, `socle`, `associated_primes`

Rank and dimension-type invariants

For an encoded finite-poset module M:

- **Restricted Hilbert function**
$$h_M(p) = \dim_{\Bbbk} M_p.$$
This is the dimension surface on the encoding poset.
- **Rank invariant**
$$\rho_M(a,b) = \operatorname{rank}\big(M(a \le b)\big), \qquad a \le b.$$
- **Rectangle signed barcode / signed measure**
On a finite grid, one may Mobius-invert the rank invariant to obtain a signed sum of rectangles. This is the multiparameter analog of turning cumulative rank data into local contributions.

Why this family matters. It is exact, finite-poset, and algebraic. It is the starting point for rectangle images, rank-from-signed-barcode recovery, and several later distances and kernels.

Slice-style invariants

Choose a monotone chain or slice λ in parameter space. Restriction gives a 1-parameter module

$$\lambda^* M,$$

and hence an ordinary barcode.

Primary objects

- `slice_barcode`, `slice_barcodes`
- fibered and projected caches for repeated slicing
- exact and approximate matching distance in 2D

Derived 1D summaries on slices

- persistence landscape
- persistence image
- persistence silhouette
- barcode entropy and barcode summary
- sliced bottleneck and sliced Wasserstein distances

Euler and signed-measure invariants

When the encoded object is a module or a cochain complex, the library also supports Euler-type data.

- **Euler surface**

$$\chi(x) = \sum_i (-1)^i \dim_{\Bbbk} C^i(x)$$

  evaluated on a chosen grid or on axes induced by the encoding.

- **Euler signed measure**

  Mobius inversion of the Euler surface on the product-of-chains grid yields a sparse signed point measure.

- **Point signed measure kernels and distances**

  These convert the sparse signed measure into a reusable low-dimensional descriptor.

Multiparameter landscapes and persistence images

There are two related image-like summary families.

- **Slice-aggregated family:** `mp_landscape`, plus the standard 1D landscape/image constructions applied slice-by-slice and aggregated.
- **Carriere-style family:** `mpp_decomposition` and `mpp_image`.

MPPI idea. The code stores a finite family of weighted line segments in the plane:

$$I_k \rightsquigarrow \{(p, q, \omega)\}_{\text{segments}}$$

and then rasterizes them into an image by Gaussian smoothing.

This gives:

- a visual descriptor,
- an inner product / kernel,
- and a distance between images.

These are especially useful for machine learning workflows where one wants a stable, fixed-size representation.

Geometry from PL encodings

For PL and box encodings, the classifier map carries actual region geometry. This is not a separate invariant family in the narrow algebraic sense, but it is a major source of geometric summaries.

Region geometry API

- size and support: `region_weights`, `region_volume`, `region_bbox`, `region_widths`, `region_diameter`
- shape: `region_centroid`, `region_aspect_ratio`, `region_principal_directions`, `region_covariance_anisotropy`, `region_covariance_eccentricity`
- boundary and adjacency: `region_adjacency`, `region_boundary_measure`, `region_perimeter`, `region_surface_area`
- inscribed / circumscribed geometry: `region_chebyshev_ball`, `region_inradius`, `region_circumradius`
- integral summaries: `region_boundary_to_volume_ratio`, `region_isoperimetric_ratio`, `region_mean_width`, `region_minkowski_functionals`

These data feed weighted invariants such as `support_measure`, `integrated_hilbert_mass`, `aspect_ratio_stats`, `interface_measure`, and `module_geometry_summary`.

Algebraic support descriptors and takeaway

The implementation also includes algebraic support summaries that sit closer to commutative algebra:

- `betti_support_measures`, `bass_support_measures`
- `minimal_primes`, `socle`, `associated_primes`
- covering summaries such as `upsets_covering` and `downsets_covering`

Takeaway.

1. The library does not implement a single "multiparameter barcode."
2. It instead keeps the finite encoded module and exposes many invariant families.
3. The right invariant depends on the question:
    - rank and Hilbert data for exact algebraic structure,
    - slices for 1D comparison matching distances, and similar
    - Euler and signed measures for compact summaries,
    - MPPI and landscapes for feature extraction,
    - PL region geometry for weighted and geometric summaries.